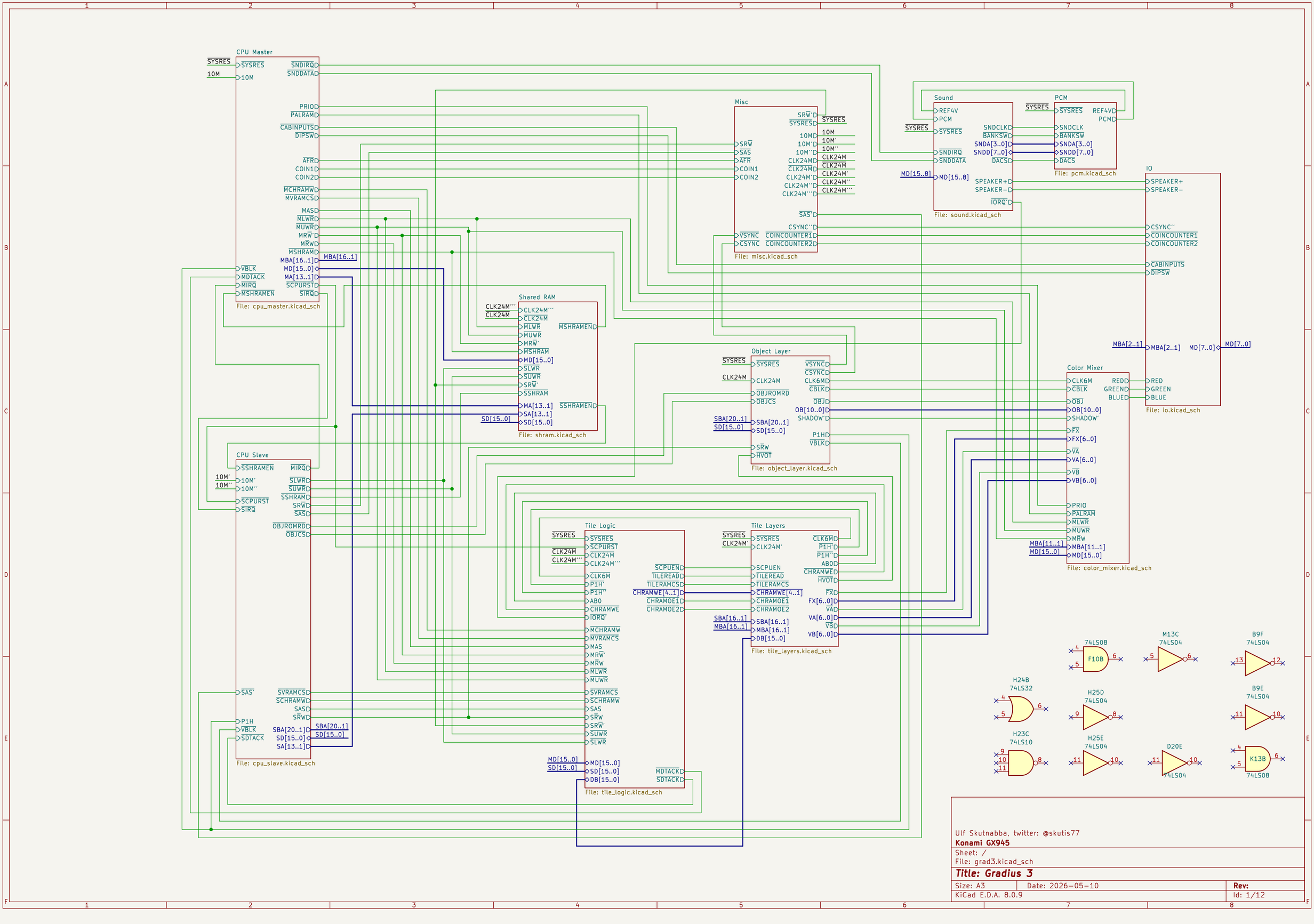


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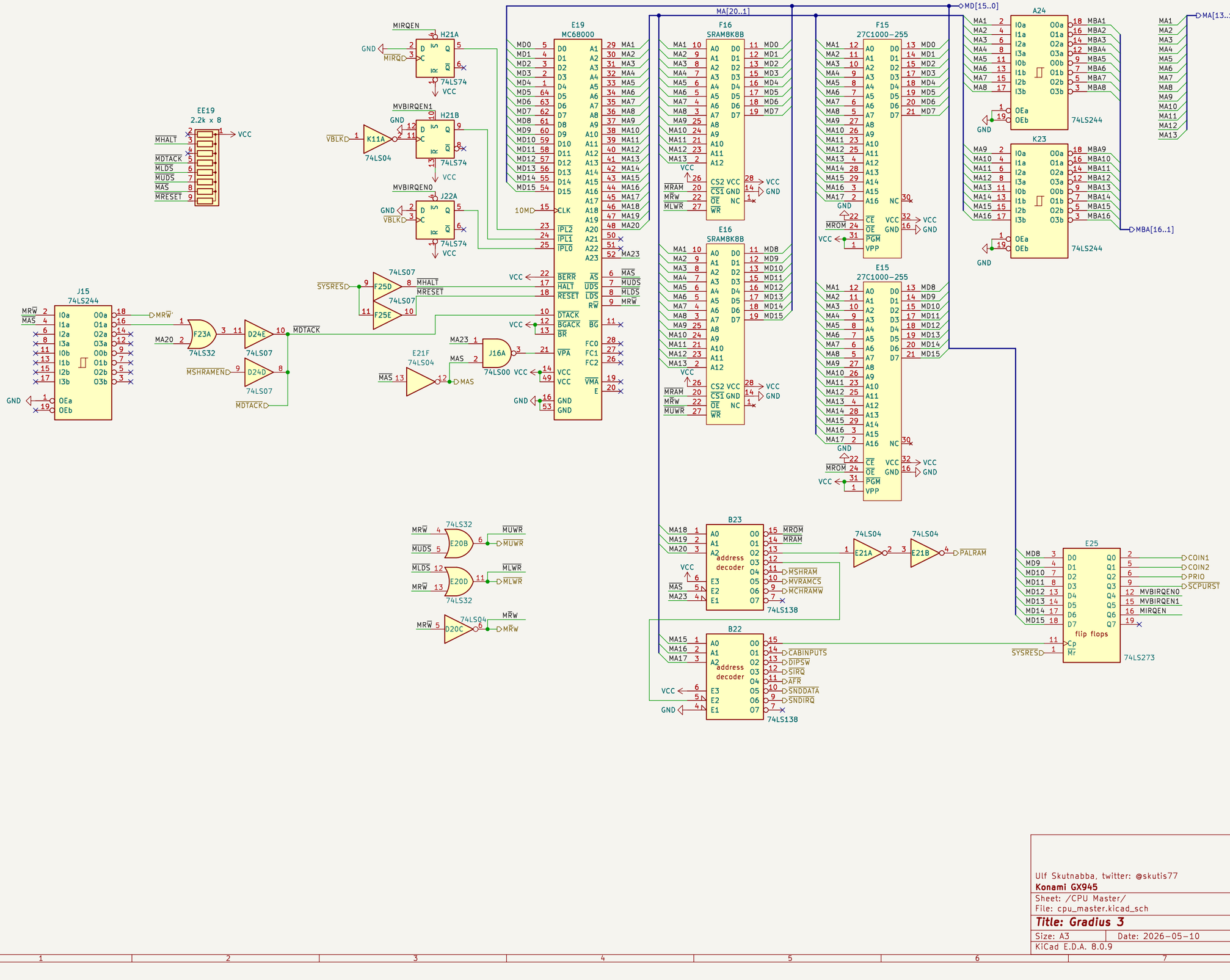
Konami GX945

Sheet: /

File: grad3.kicad_sch

Title: Gradius 3

Size: A3	Date: 2026-05-10	**Rev:**
KiCad E.D.A. 8.0.9		Id: 1/12



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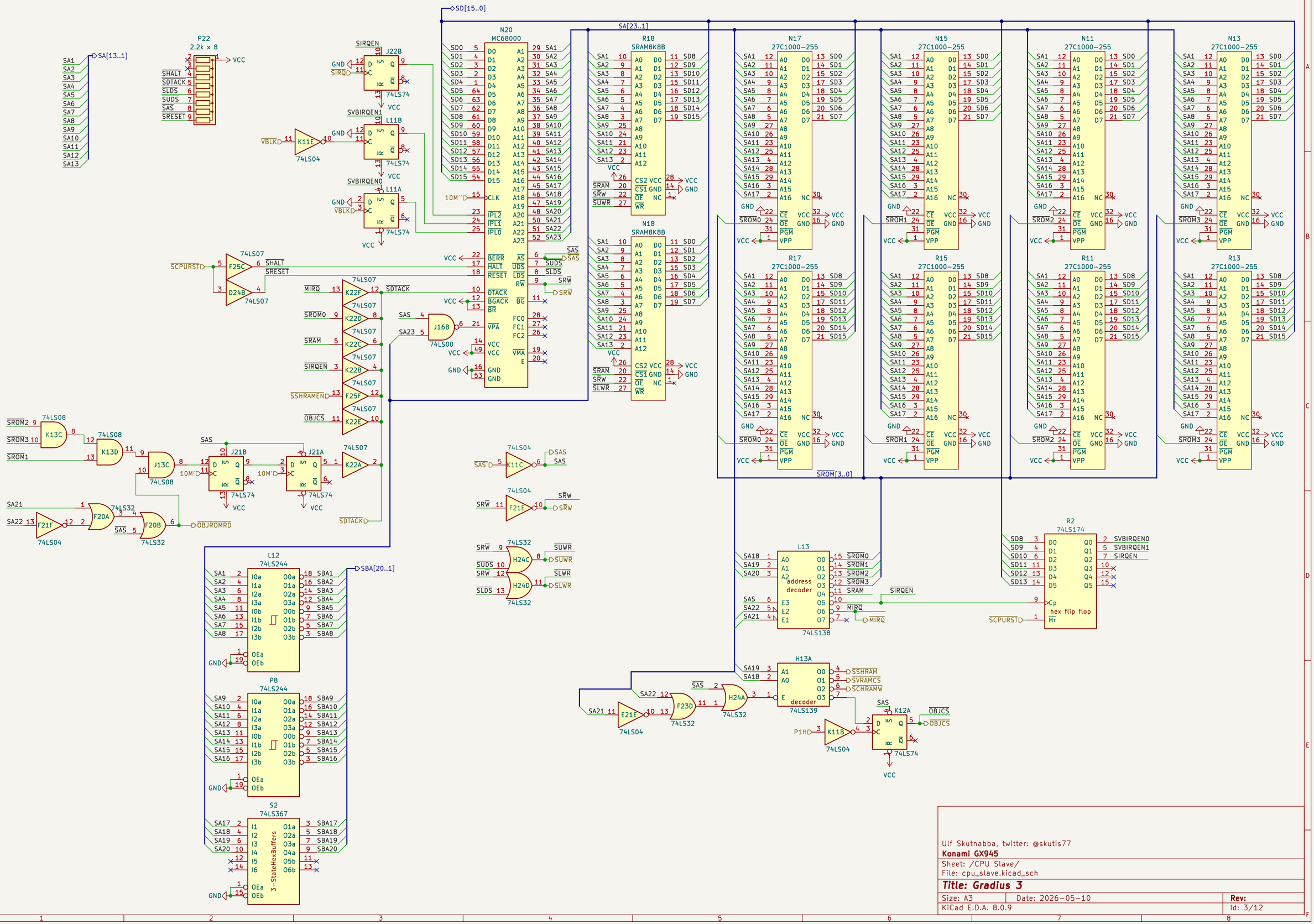
Konami GX945

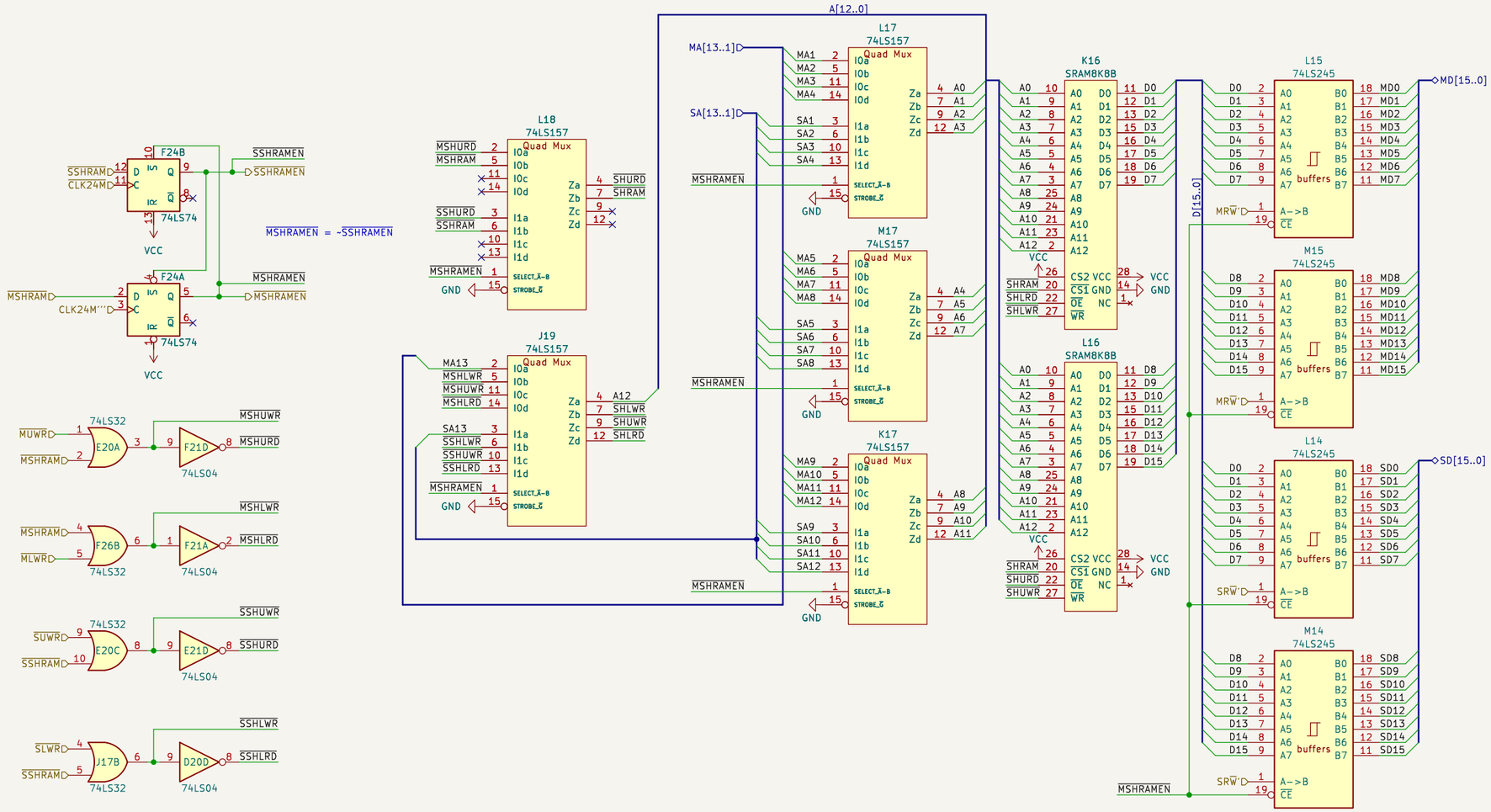
Sheet: /CPU Master/

File: cpu_master.kicad_sch

Title: Gradius 3

Size: A3	Date: 2026-05-10	**Rev:**
KiCad E.D.A. 8.0.9		Id: 2/12


SD[15..0]
SA[23..1]
SA[13..1]
N20
MC68000
R18
SRAM8K8B
N18
SRAM8K8B
N17
27C1000-255
R17
27C1000-255
N15
27C1000-255
R15
27C1000-255
N11
27C1000-255
R11
27C1000-255
N13
27C1000-255
R13
27C1000-255
SROM[3..0]
P22
2.2k x 8
VCC
SHALT
SDTACK
SLDS
SUDS
SAS
SRESET
SIRQEN
J22B
74LS74
SVBIRQEN1
L11B
SVBIRQEN0
L11A
K11E
74LS04
VBLK
GND
74LS07
F25C
D24B
SCPURST
SHALT
SRESET
MIRQ
K22F
K22D
K22C
K22B
F25F
K22E
SROM0
SRAM
SIRQEN
SSHRAMEN
OBJCS
SDTACK
J16B
74LS00
SAS
SA23
10M'
BERR
HALT
RESET
DTACK
BGACK
BR
AS
UDS
LDS
RW
BG
VPA
FCO
FC1
FC2
VMA
E
SRW
SUDS
SLDS
74LS08
K13C
K13D
J13C
J21B
J21A
K22A
SROM2
SROM3
SROM1
SA21
SA22
F21F
F20A
F20B
74LS32
OBJROMRD
74LS04
K11C
F21E
H24C
H24D
SUWR
SLWR
L12
74LS244
P8
74LS244
S2
74LS367
3-StateHexBuffers
SBA[20..1]
L13
address decoder
74LS138
H13A
decoder
74LS139
F23D
H24A
E21E
K12A
K11B
P1HD
SSHRAM
SVRAMCS
SCHRAMW
R2
74LS174
hex flip flop
SVBIRQEN0
SVBIRQEN1
SIRQEN
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Konami GX945
Sheet: /CPU Slave/
File: cpu_slave.kicad_sch
Title: Gradius 3
Size: A3
Date: 2026-05-10
KiCad E.D.A. 8.0.9
Rev:
Id: 3/12



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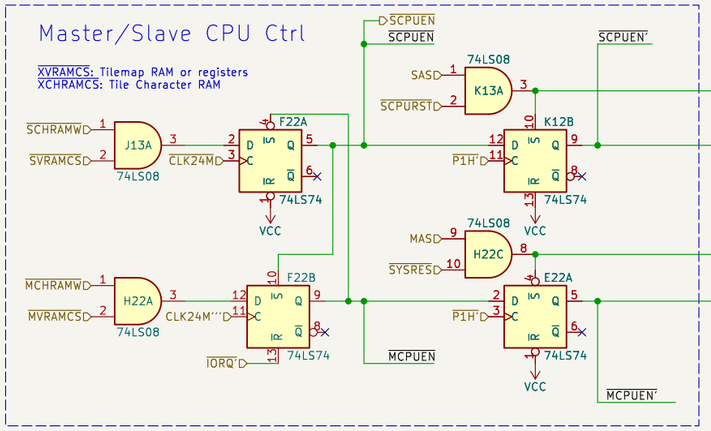
Sheet: /Shared RAM/
File: shram.kicad_sch

Title: Gradius 3

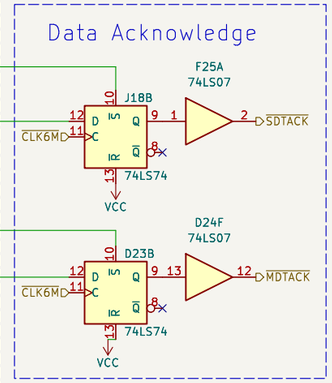
Size: A3	Date: 2026-05-10	**Rev:**
KiCad E.D.A. 8.0.9		Id: 4/12

Master/Slave CPU Ctrl

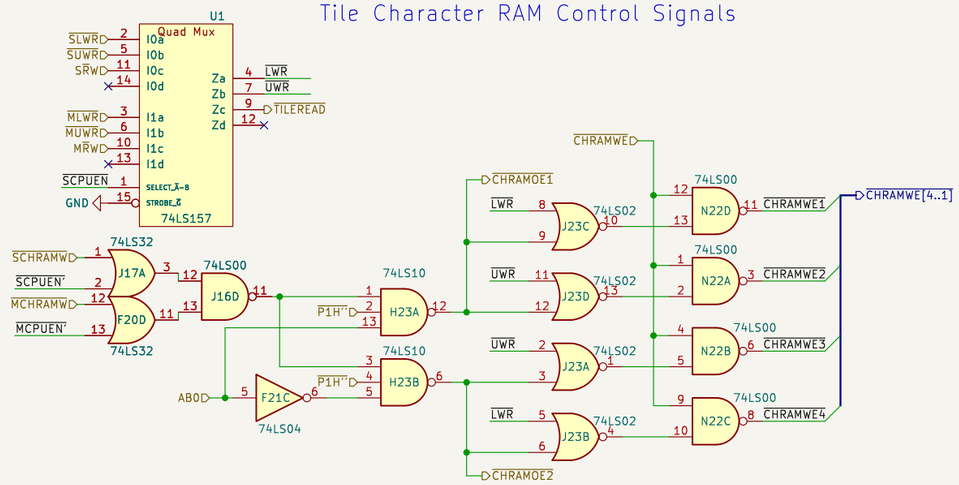
XVRAMCS: Tilemap RAM or registers
XCHRAMCS: Tile Character RAM



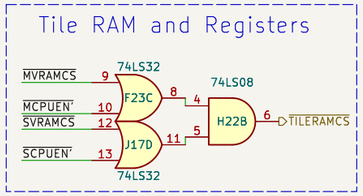
Data Acknowledge



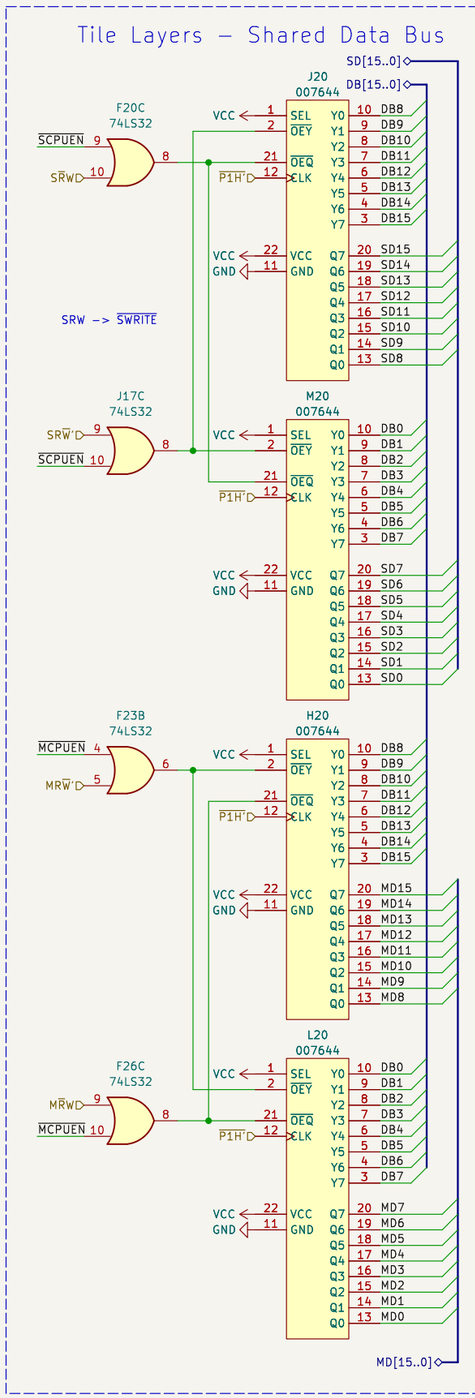
Tile Character RAM Control Signals



Tile RAM and Registers



Tile Layers — Shared Data Bus



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Sheet: /Tile Logic/
File: tile_logic.kicad_sch

Title: Gradius 3

Size: A3	Date: 2026-05-10	**Rev:**
KiCad E.D.A. 8.0.9		Id: 5/12

Character code RAM

TILERAMCS can write and read both to tilemap RAM and character RAM

Tilemaps RAM: Fix, Layer 1 and 2

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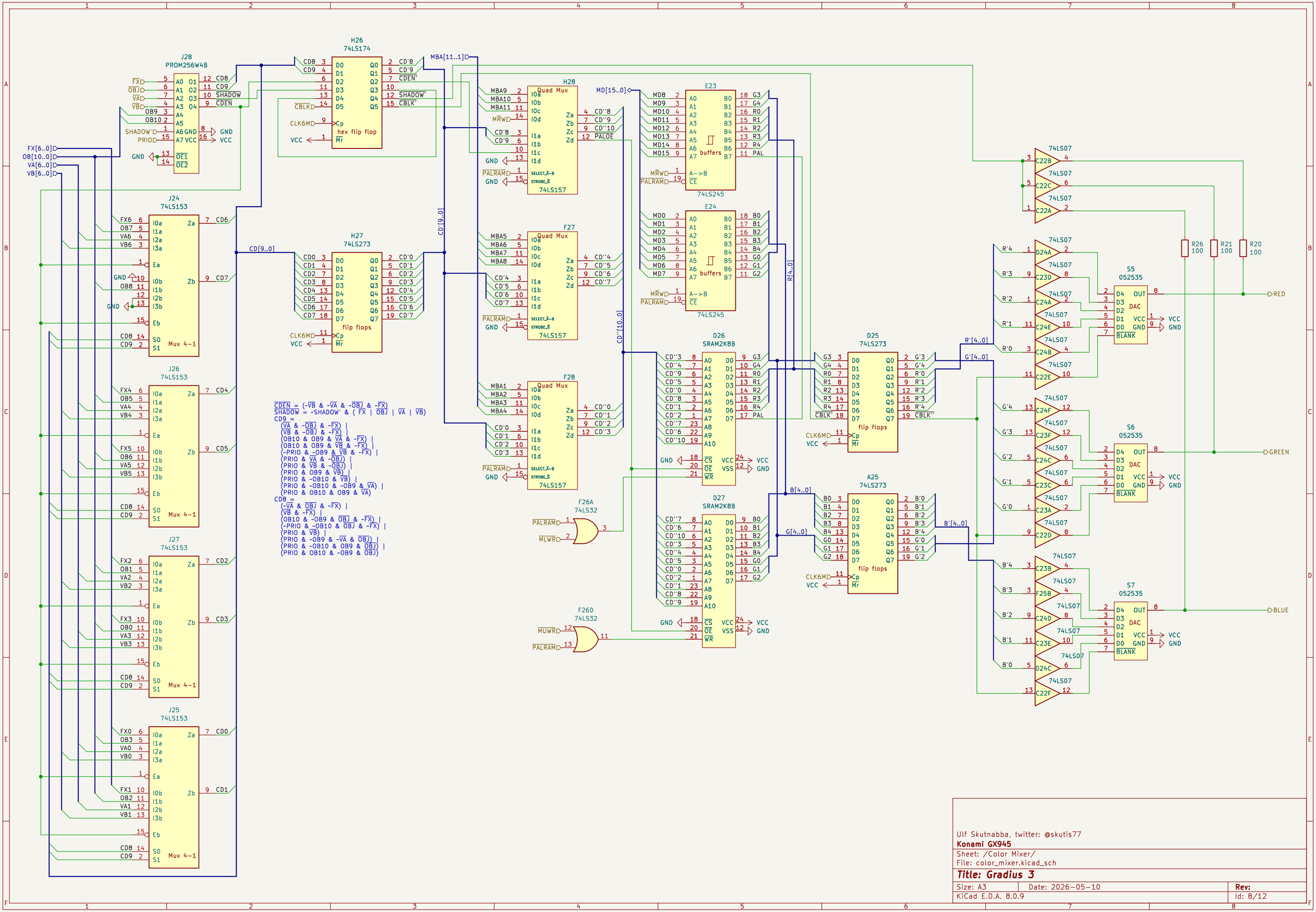
Sheet: /Tile Layers/

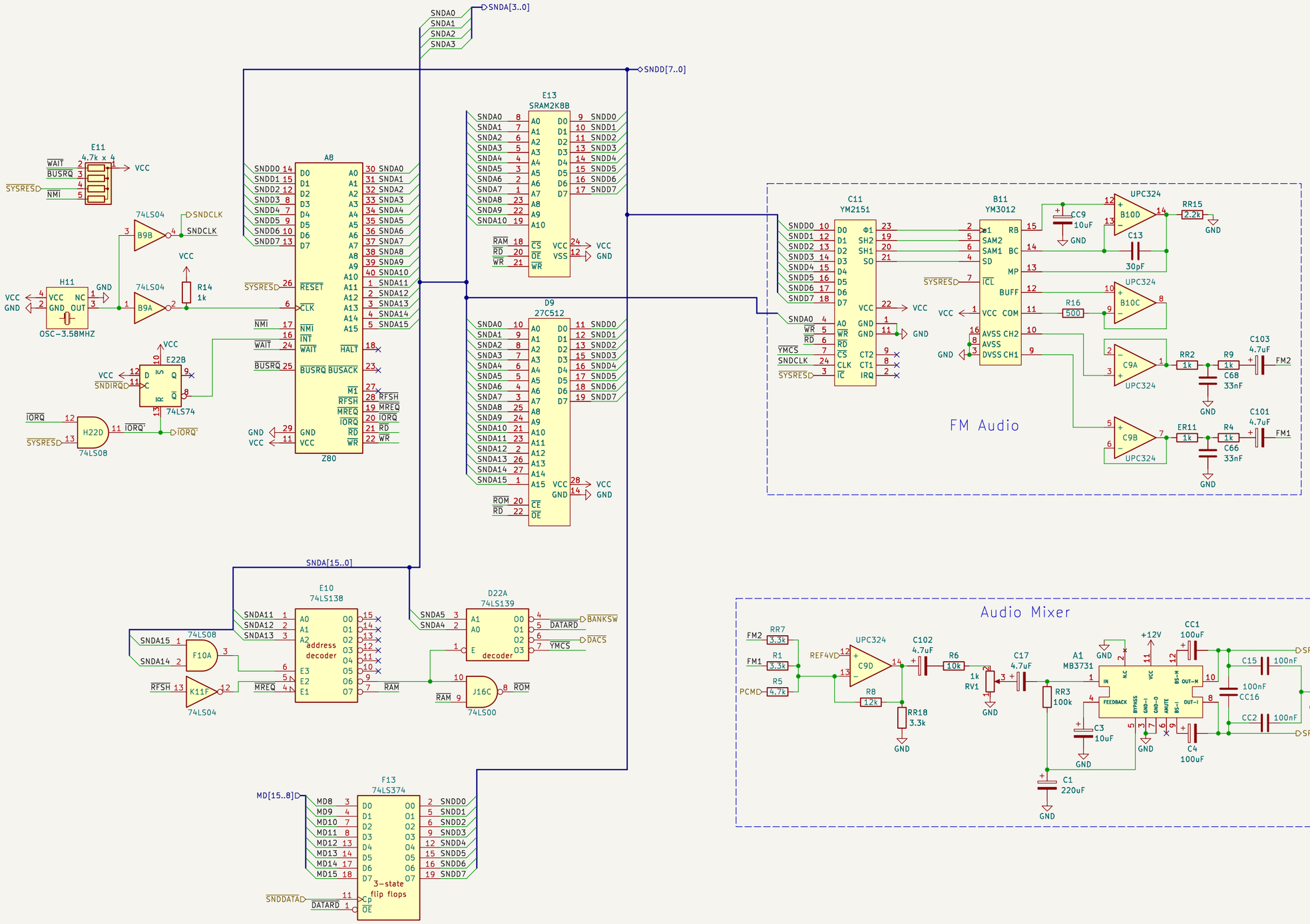
File: tile_layers.kicad_sch

Title: Gradius 3

Size: A3	Date: 2026-05-10	**Rev:**
KiCad E.D.A. 8.0.9		Id: 6/12

R7
051960
SBA[20..1]
CA[17..0]
OBJROMRD
OC[7..1]
HP[8..0]
OA[9..0]
OD[7..0]
{OBJCTRL}
P7
74LS157
Quad Mux
P5
74LS157
Quad Mux
P4
74LS157
Quad Mux
P3
74LS157
Quad Mux
P2
74LS157
Quad Mux
H6
K6
M6
E6
H8
K8
M8
E8
27C1000
R3
SRAM2K8B
H13B
74LS139
decoder
A[18..0]
OROM[1..0]
SD[15..0]
CD[37..0]
H10
74LS244
J10
74LS244
L10
74LS244
M10
74LS244
P6C
74LS32
P6D
74LS32
P6A
74LS32
P6B
74LS32
K11D
74LS04
J13D
74LS08
J13B
74LS08
F10C
74LS08
F10D
74LS08
Goes to optional ROM Modules
Not used on gradius 3
H3
23C4100
L3
23C4100
D4
051937
OB[10..0]
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Konami GX945
Sheet: /Object Layer/
File: object_layer.kicad_sch
Title: Gradius 3
Size: A3
Date: 2026-05-10
Rev:
KiCad E.D.A. 8.0.9
Id: 7/12


J28
PROM256W4B
H26
74LS174
MBA[11..1]
MD[15..0]
H28
Quad Mux
E23
74LS245
E24
74LS245
J24
74LS153
H27
74LS273
F27
Quad Mux
F28
Quad Mux
D26
SRAM2K8B
D25
74LS273
A25
74LS273
J26
74LS153
J27
74LS153
J25
74LS153
F26A
74LS32
F26D
74LS32
D27
SRAM2K8B
S5
052535
S6
052535
S7
052535
RED
GREEN
BLUE
FX[6..0]
OB[10..0]
VA[6..0]
VB[6..0]
CD[9..0]
CD''[9..0]
CD''[10..0]
R[4..0]
B[4..0]
G[4..0]
R'[4..0]
G'[4..0]
B'[4..0]
CDEN = (~VB & ~VA & ~OBJ & ~FX)
SHADOW = ~SHADOW' & (FX | OBJ | VA | VB)
CD9 =
(VA & ~OBJ & ~FX) |
(VB & ~OBJ & ~FX) |
(OB10 & OB9 & VA & ~FX) |
(OB10 & OB9 & VB & ~FX) |
(~PRIO & ~OB9 & VB & ~FX) |
(PRIO & VA & ~OBJ) |
(PRIO & VB & ~OBJ) |
(PRIO & OB9 & VB) |
(PRIO & ~OB10 & VB) |
(PRIO & ~OB10 & ~OB9 & VA) |
(PRIO & OB10 & OB9 & VA)
CD8 =
(~VA & OBJ & ~FX) |
(VB & ~FX) |
(OB10 & ~OB9 & OBJ & ~FX) |
(~PRIO & ~OB10 & OBJ & ~FX) |
(PRIO & VB) |
(PRIO & ~OB9 & ~VA & OBJ) |
(PRIO & ~OB10 & OB9 & OBJ) |
(PRIO & OB10 & ~OB9 & OBJ)
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Konami GX945
Sheet: /Color Mixer/
File: color_mixer.kicad_sch
Title: Gradius 3
Size: A3
Date: 2026-05-10
Rev:
KiCad E.D.A. 8.0.9
Id: 8/12



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Sheet: /Sound/

File: sound.kicad_sch

Title: Gradius 3

Size: A3	Date: 2026-05-10	**Rev:**
KiCad E.D.A. 8.0.9		Id: 9/12

B15 and B18 do not follow the standard Mask ROM behaviour. The BYTE mode select must be selected from factory and pin 31 is probably not connected.

BANK A

Mounted on all releases except (Japan, program code S, split)

B15
23C4100

BANK B

Not mounted on any releases

B18
23C4100

Mounted on Gradius III (Japan, program code S, split)

C14
27C1000

C18
27C1000

C16
27C1000

C20
27C1000

PCM ROM Data Bank Select

D21
74LS174

B21
74LS157
Quad Mux

C21
74LS174

74LS04

D22B
74LS139
decoder

SNDD[7..0]

BANKSW

SYSRES

SNDCLK4

SNDCLK2

DABNK0

DABNK1

DABNKA0

DABNKA1

DABNKB0

DABNKB1

SELECT_A-B

STROBE_E

SNDCLK
3.579545MHz

SNDCLK2
1.789773MHz

SNDCLK4
0.894886MHz

SNDA[3..0]

SNDA0

74LS04

C2
007232

SNDCLK

DACS

SYSRES

VCC

GND

AS[16..0]

DS[7..0]

74LS74

D7
74LS273
flip flops

D10
4.7k x 8

Q[0..7]

EE6
4066

D8
KONAMI_007341

C9C
UPC324

C63
4.7uF

PCM

REF4V

R10
10k

GND B3

007340
A3

RR13
20k

RR12
20k

C10
470pF

C105
470pF

C104
4.7uF

C100
4.7uF

B7

B8

4.7k x 4

UPC324

B10B

B10A

+12V

R19
3.3k

RR17
3.3k

CC21
4.7uF

CC14
4.7uF

C64
4.7uF

D6
4066

DD7
KONAMI_007341

B10E
UPC324

C9E
UPC324

The values to the right are the capacitors on the PCB. C10X are not visible on the PCB. They are values set to ease debugging.

C104
C103
C102
C10
C101
C100
C65
C67
C105

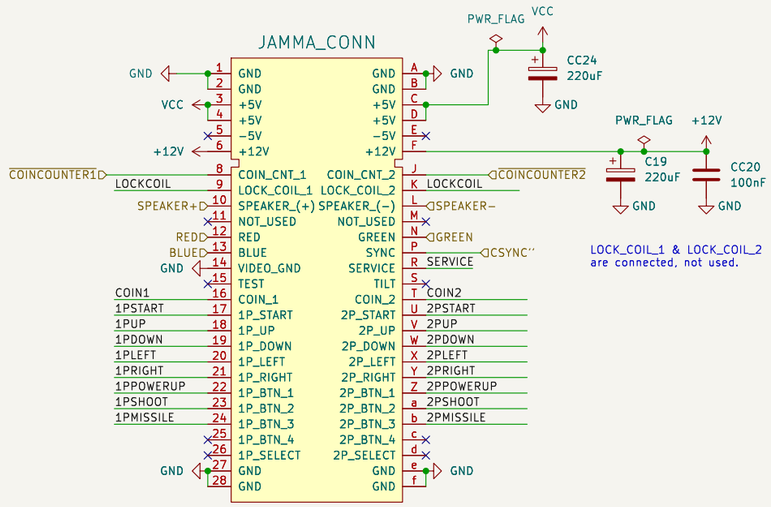
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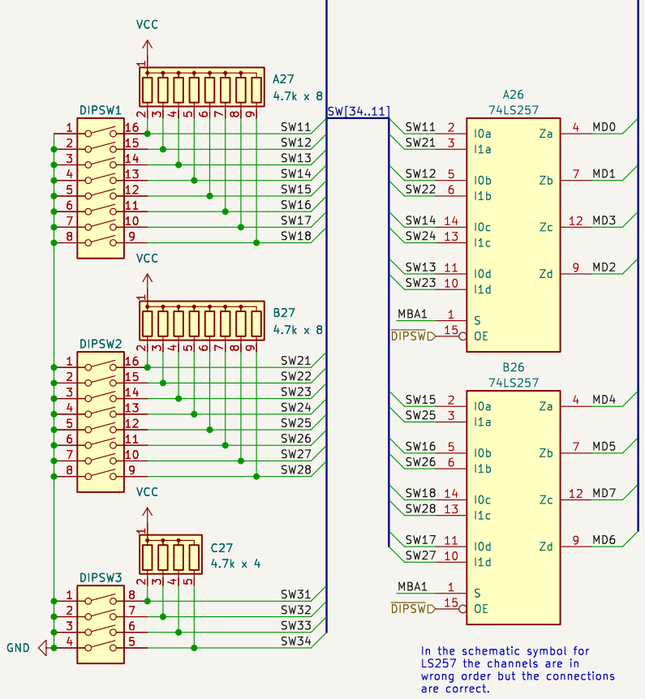
Konami GX945

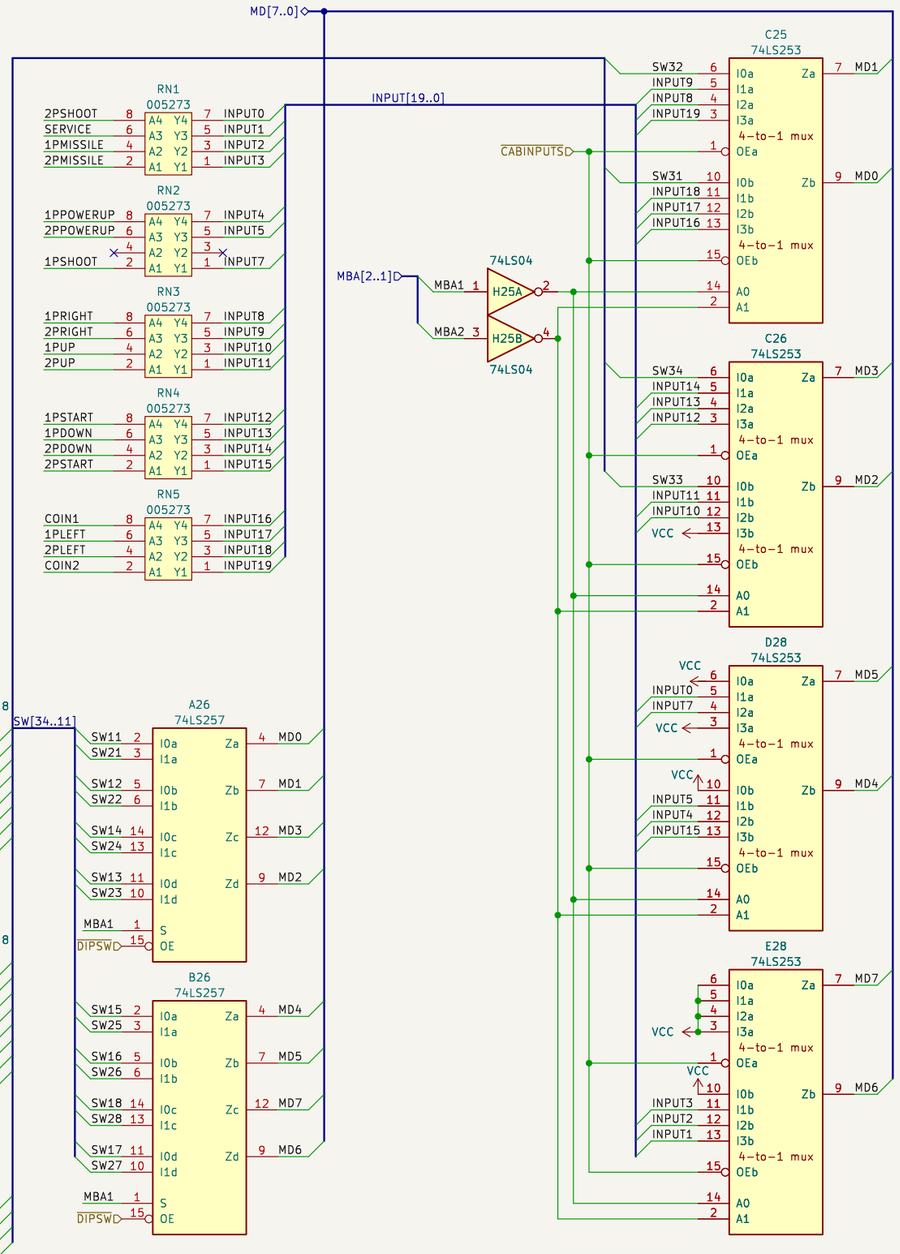
Sheet: /PCM/
File: pcm.kicad_sch

Title: Gradius 3

Size: A3	Date: 2026-05-10	**Rev:**
KiCad E.D.A. 8.0.9		Id: 10/12







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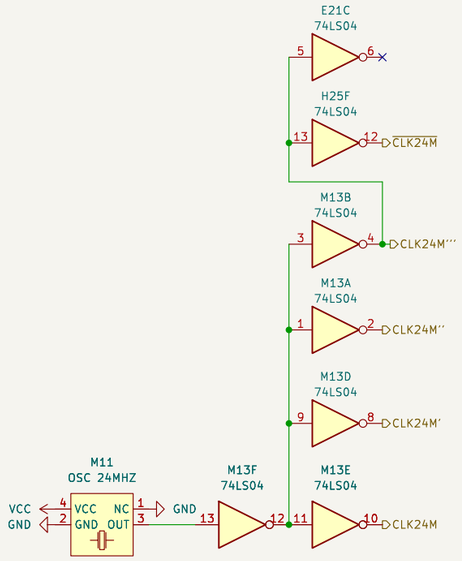
Konami GX945

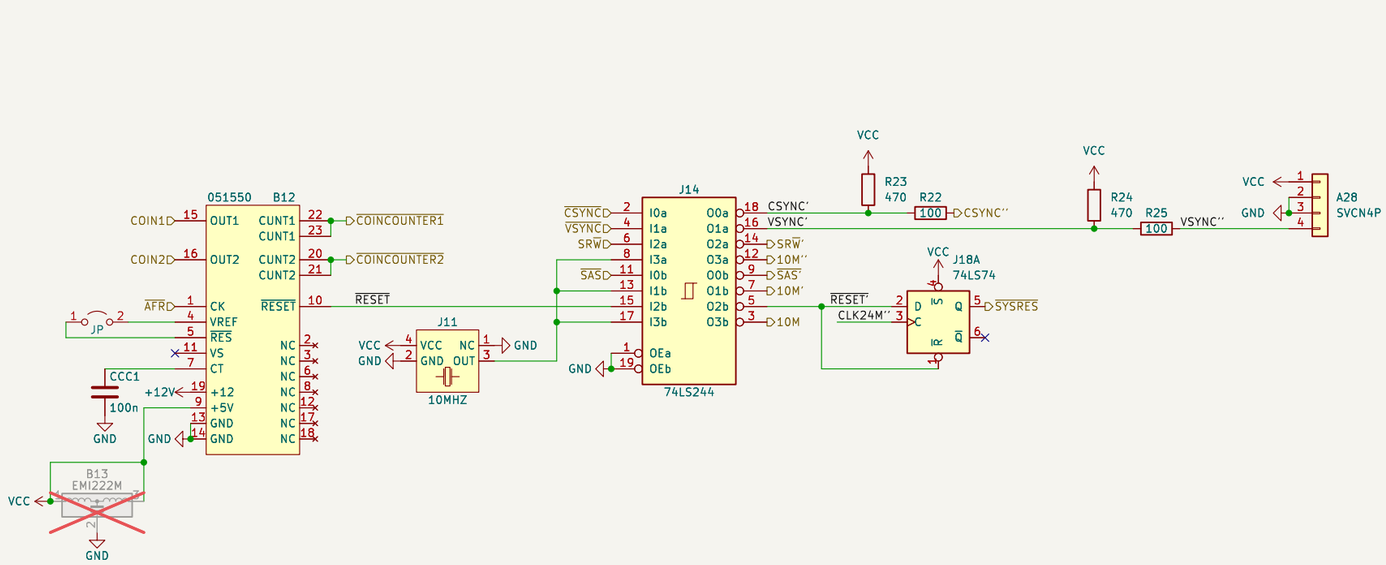
Sheet: /IO/

File: io.kicad_sch

Title: Gradius 3

Size: A3	Date: 2026-05-10	**Rev:**
KiCad E.D.A. 8.0.9		Id: 11/12





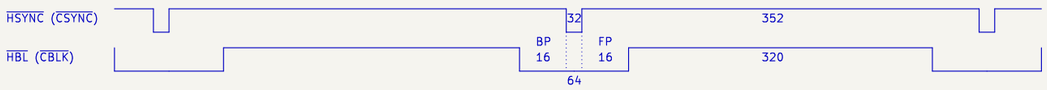
Horizontal and vertical synch timing diagrams

The pixel clock is derived from the 24MHz oscillator.
Pixel clock OVCK: f = 24MHz / 4 = 6MHz

The numbers in the HSYNC and HBL diagram are pixel clock cycles.
All edges are synchronised to the rising edge of the pixel clock.

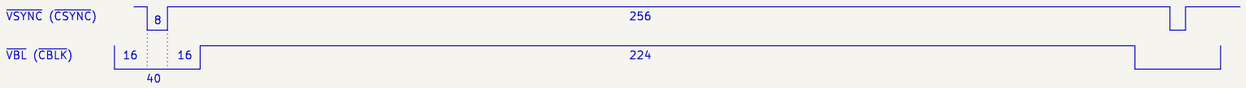
The signals have been measured at the output of the
graphic chips.

If horizontal blanking is measured at the RGB DACs, the blanking
is delayed 2 pixel clocks relative to composite sync. This
gives BP = 14 and FP = 18.



HSYNC and HBL
Frequency f = 6MHz / 384 = 15.625kHz.
Period T = 1/f = 64us.

The numbers in the VSYNC and VBL diagram are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



VSYNC and VBL:
Frequency f = 15.625kHz / 264 = 59.1856Hz
Period T = 1/f = 1 / 59.1856Hz = 16.896ms

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Konami GX945

Sheet: /Misc/
File: misc.kicad_sch

Title: Gradius 3

Size: A3	Date: 2026-05-10	**Rev:**
KiCad E.D.A. 8.0.9		Id: 12/12